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Keywords:

Flexure – Hall sensor –
Displacement characteri-
zation

Experimental methods for testing materials using rotational forces

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Abstract

Unlike conventional material testing systems, which often impose displacement and measure the resulting force for a single specimen, a rotating test rig inherently imposes forces that can be exploited for high-throughput material testing. Accurate displacement measurements within the rotating frame are therefore required to characterize the mechanical response of the tested structures.

To address this challenge, two complementary displacement measurement techniques were developed and evaluated. A Hall-effect sensor combined with a permanent magnet was used as an embedded sensing solution, while an optical tracking method based on camera imaging provided an independent reference measurement. Experiments were conducted on flexure-guided translation stages mounted on a rotating platform and subjected to centrifugal loading.

Both measurement techniques successfully captured the displacement evolution of the moving carriage. Larger displacements were obtained for longer flexures and higher rotational velocities, consistent with the increase in centrifugal loading and the reduction in structural stiffness. A direct comparison between the Hall-effect sensor and camera measurements showed good agreement in terms of displacement amplitude and overall profile shape. To enable this comparison, the experimental profiles were normalized and reconstructed using the theoretical motor cycle duration associated with each target velocity.

The camera-based measurements provided a reliable reference, while the Hall-effect sensor successfully reproduced the main displacement trends. The principal limitation of the Hall-effect sensor approach lies in the calibration procedure, which remains dependent on the specific sensor-magnet assembly and was established from a limited number of calibration points. Nevertheless, the results demonstrate the feasibility of embedded Hall-effect sensing for displacement monitoring in rotating systems and provide a foundation for future testing applications.

Introduction

When a mass m is subjected to rotation, inertial effects generate apparent forces that can be used to mechanically load a structure. By controlling the rotational velocity, it becomes possible to generate a repeatable tensile loading without relying on conventional actuators or testing machines. Such an approach is particularly attractive for fatigue applications, where long-duration testing and the simultaneous loading of multiple specimens are desirable.

In a rotating reference frame, three apparent forces arise: the centrifugal force $F^\Omega = m\Omega^2 R$, acting radially outward; the Euler force $F^\epsilon = -m\dot{\Omega}R$, associated with angular acceleration; and the Coriolis force $F^C = -2m\Omega \times v_r$, which appears during radial motion. In the present design, parallel leaf-spring flexures constrain all degrees of freedom except radial translation. As a result, centrifugal loading dominates the specimen response, while Euler and Coriolis effects remain negligible.

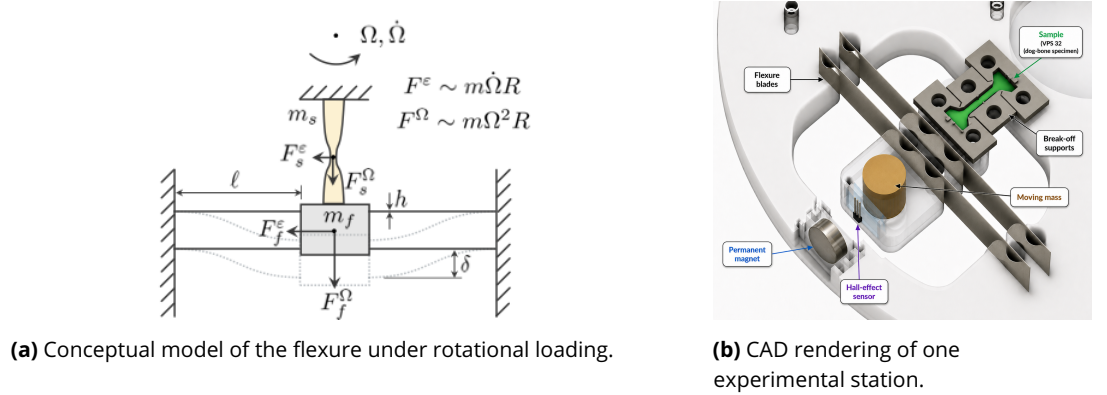


Figure 1. Experimental implementation and measurement principle of one testing station. (a) Schematic representation of the loading mechanism. The centrifugal force F^Ω generates a radial tensile load on the specimen while the flexure blades guide the motion of the moving assembly. (b) CAD rendering of the testing station showing the flexure blades, moving mass, Hall-effect sensor, permanent magnet, dog-bone specimen, and break-off supports. Displacement is measured through variations in the magnetic field detected by the Hall-effect sensor.

To exploit centrifugal loading for material testing, a rotating rig was developed in which the specimen is attached to a carriage guided by parallel leaf-spring flexures (Figure 1a). Unlike conventional testing machines that impose displacement, the rotating rig applies force through the rotation speed. Consequently, displacement measurements are required to characterize the specimen response.

The use of flexure-guided translation stages is well established in precision engineering applications due to their ability to provide highly repeatable motion without friction, backlash, or wear. In particular, zero-force rectilinear flexure mechanisms have demonstrated the capability to generate nearly pure translational motion while minimizing parasitic forces and unwanted degrees of freedom [3]. In the present design, the parallel-flexure arrangement constrains the carriage to radial motion, such that the measured displacement primarily reflects the deformation of the specimen and the flexure itself.

For small displacements, the flexure response can be approximated using Euler–Bernoulli beam theory,

$$\delta = \frac{F \ell^3}{3EI} \quad (1)$$

where E is the Young's modulus of the flexure material, I is the second moment of area of the flexure cross-section, and ℓ is the effective flexure half-span. This expression provides a first-order analytical estimate of the displacement and remains valid when beam rotations are small.

For larger displacements, geometric nonlinearities become significant and the force-displacement relationship deviates from the linear Euler–Bernoulli prediction. In this regime, the effective stiffness of the flexure increases due to large beam rotations. A nonlinear formulation proposed by Henein [4] can be used to describe this behaviour:

$$F(\delta) = \frac{6 b h E \delta^4}{3 \delta I^3 - \sqrt{6} h I^3 \coth\left(\sqrt{\frac{3}{2}} \frac{\delta}{h}\right)} \quad (2)$$

where b , h , and I denote the width, thickness, and length of the flexure blade, respectively. This expression accounts for large-deflection effects and predicts the nonlinear stiffening behaviour observed as displacement increases. Such geometric nonlinearities are commonly encountered in compliant mechanisms and flexure-based systems and have been extensively investigated in the literature [5].

Several non-contact acquisition methods were investigated to monitor displacement and detect rupture during rotation. These include (I) a Hall-effect sensor recovering carriage displacement from magnetic flux variations, (II) optical imaging using camera tracking, and (III) monitoring of the electrical continuity of a conductive surface coating applied to the specimen. The latter method provides a simple means of rupture detection but does not provide continuous displacement information and was therefore not investigated further.

The Hall-effect sensor and optical imaging methods were retained and experimentally compared regarding accuracy, robustness, and suitability for long-duration fatigue testing in a rotating environment.

The objective of this project is to characterize the mechanical response of a flexure-guided centrifugal testing platform and to evaluate displacement measurement techniques suitable for operation in a rotating environment. Particular attention is given to the comparison between Hall-effect sensing and camera-based tracking, as well as to the influence of flexure stiffness on the displacement response generated by centrifugal loading. The long-term objective is to provide a reliable measurement framework for future fatigue testing and rupture detection applications.

Methods

Experimental setup

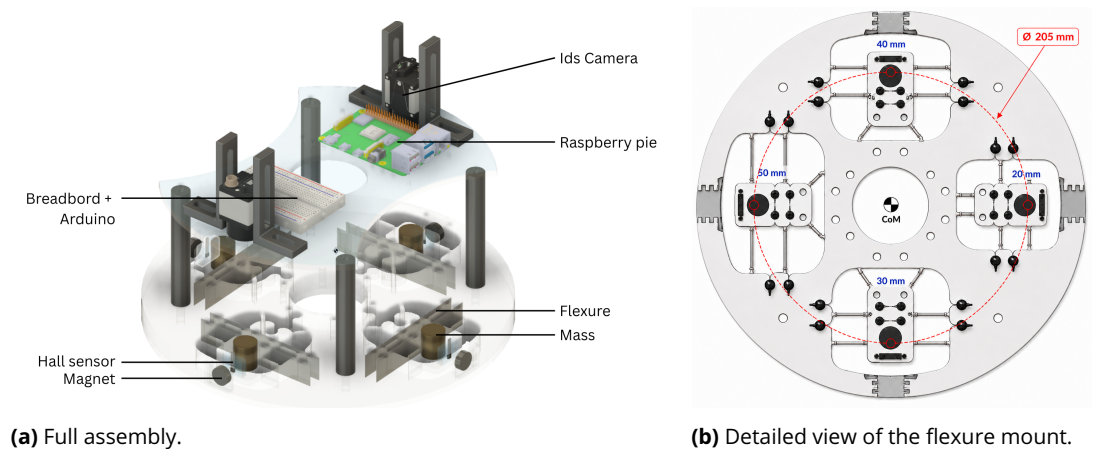


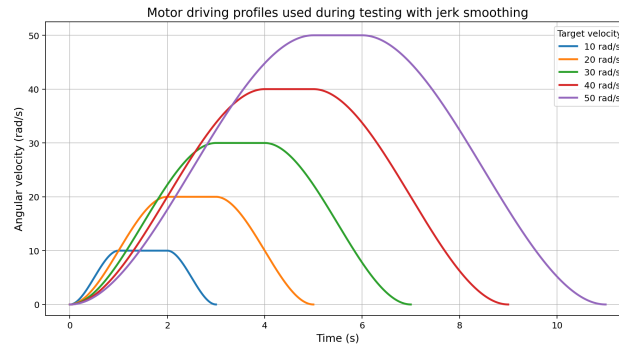
Figure 2. CAD rendering of the experimental apparatus. Panel (a) shows the complete rotating assembly with the four testing stations, while panel (b) highlights the flexure guiding system and carriage mounting region. The red dashed line represents the distance between the rotation axis and the centre of mass of the moving assembly. This distance is identical for all testing stations, ensuring that the same centrifugal force is generated for a given angular velocity.

The experimental apparatus shown in Figure 2a consists of a rotating acrylic structure carrying four identical testing stations mounted symmetrically around the rotation axis. The complete assembly is mounted directly onto the shaft of an ETEL RTMBi140-050-3RBS direct-drive torque motor [2] through a dedicated fixation interface.

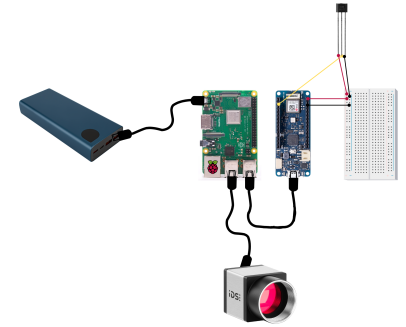
The structure was manufactured from three laser-cut acrylic plates to ensure good dimensional accuracy and repeatability. Vertical aluminium rods support an electronics plate carrying the Arduino microcontroller, Raspberry Pi, external battery, camera support, and acquisition hardware. Hall-effect sensors and permanent magnets are integrated into removable laser-cut acrylic holders to facilitate positioning and calibration.

The four testing stations differ only in the flexure length, with values of 20 mm, 30 mm, 40 mm and 50 mm. This configuration enables the influence of flexure stiffness on the displacement response to be investigated while maintaining identical loading conditions. All flexures have a constant thickness of 0.20 mm and a width of 13.00 mm. The centres of mass of the four testing stations are located on a common pitch circle of 205 mm diameter, ensuring that each station have the same centrifugal loading for a given angular velocity.

A detailed view of one testing station is shown in fig. 2b. The specimen, either acrylic or VPS 32 resin, is mounted on a dedicated fixture mechanically fastened to the carriage through a screw assembly. Break-off supports are incorporated into the specimen geometry to facilitate fabrication and handling prior to testing.



(a) Motor velocity profiles.



(b) Data acquisition architecture.

Figure 3. Experimental control and acquisition system. Panel (a) shows the theoretical motor velocity profiles used during testing. Smooth jerk-limited acceleration and deceleration ramps were applied to reach target angular velocities ranging from 10 rad s^{-1} to 50 rad s^{-1} . These profiles define the reference loading sequence applied during the experiments. Panel (b) presents the electrical architecture used for data acquisition, camera operation, and storage of experimental measurements.

The rotational speed was controlled using the jerk-limited motion profiles shown in Figure 3a. Five target angular velocities were investigated: 10 rad s^{-1} , 20 rad s^{-1} , 30 rad s^{-1} , 40 rad s^{-1} and 50 rad s^{-1} . Each profile consisted of an acceleration phase, a constant-speed plateau, and a deceleration phase. The profile duration was adjusted according to the target velocity while maintaining a comparable loading sequence between experiments.

Displacement measurement via Hall-effect sensor

Displacement measurements were performed using a linear Hall-effect sensor (Texas Instruments DRV5013) [7] coupled with a cylindrical neodymium permanent magnet (Eclipse N829, 15 mm diameter) [1]. The Hall-effect sensor was mounted on the fixed structure while the permanent magnet was attached to the moving carriage. Any radial displacement therefore modified the sensor-to-magnet distance and produced a variation in the sensor output.

A preliminary calibration was carried out to establish the relationship between the Hall-effect sensor output and carriage displacement. The resulting calibration curve is provided in Appendix. Although the calibration was used to estimate displacement amplitudes, the present work primarily focuses on the analysis of the raw Hall-effect sensor response and its comparison with camera-based measurements.

To avoid signal saturation at large displacements, the setup incorporated three discrete magnet positions corresponding to initial sensor-to-magnet distances of 15 mm, 20 mm and 25 mm. The appropriate position was selected according to the expected displacement range of each experiment.

The Hall-effect sensor, permanent magnet, and acquisition electronics were mounted directly on the rotating assembly. Sensor data were acquired using an Arduino MKR WiFi 1010 and transmitted to a Raspberry Pi for storage and experiment monitoring as seen in Figure 3b.

Displacement measurement via imaging

A second displacement measurement method was developed using optical tracking of the moving carriage. The system is based on an IDS uEye CP industrial camera [6] equipped with an 8 mm lens and mounted directly on the rotating frame of the experimental apparatus.

The camera recorded video sequences at a resolution of 1440×1080 pixels and a frame rate of 60 fps. Automatic exposure and gain control were disabled, and the exposure time was fixed at $5000 \mu\text{s}$ to ensure consistent image quality throughout the experiments.

A 2 mm black circular marker attached to the moving carriage was used as a tracking target. Illumination was provided by an external light source positioned around the setup.

The recorded videos were processed offline using a dedicated Python tracking algorithm. The position of the marker was extracted frame-by-frame to reconstruct the carriage displacement as a function of time. In addition to displacement measurements, the imaging system provided direct visualisation of the flexure behaviour and enabled observation of vibrations, motion stability, and specimen rupture during testing.

Results and discussion

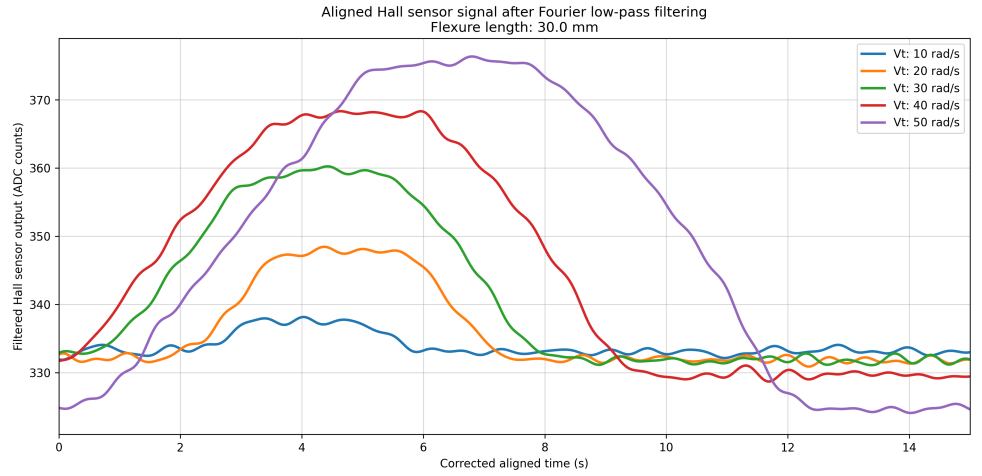
Hall-effect sensor raw measurements

The filtered Hall-effect sensor profiles were obtained for the 30 mm and 50 mm flexure configurations tested without a specimen. In order to focus on the dynamic response of the system, the analysis is performed directly on the filtered raw Hall-effect sensor output expressed in ADC counts. This approach avoids introducing additional uncertainty associated with the displacement calibration and allows the temporal evolution of the sensor signal to be compared directly with the imposed motor profile.

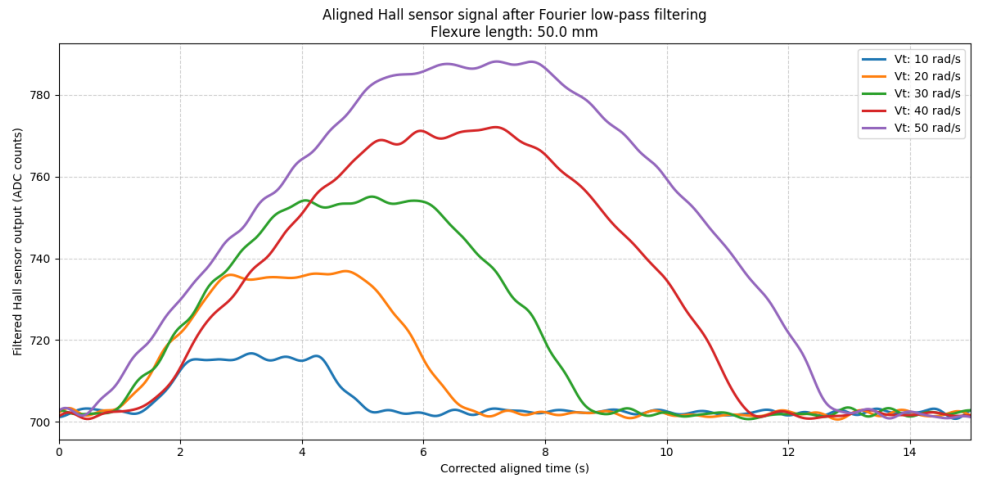
The raw signals were temporally aligned from the beginning of the significant variation from the initial baseline, and long constant regions were removed. A Fourier-based low-pass filtering was then applied to reduce high-frequency noise while preserving the global shape of the measured profiles,

$$s(t) = a_0 + \sum_{n=1}^N \left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right) \right], \quad (3)$$

where T is the duration of the analysed signal and N is the number of retained harmonics.



(a) 30 mm flexure configuration.



(b) 50 mm flexure configuration.

Figure 4. Filtered raw Hall-effect sensor outputs after temporal alignment and Fourier-based low-pass filtering, expressed in ADC counts, for rotational velocities ranging from 10 rad s^{-1} to 50 rad s^{-1} .

The shape of the Hall-effect sensor signals in Figure 4 is consistent with the motion imposed by the motor driving profiles shown in Figure 3a. The signal increases progressively during acceleration, reaches a plateau region, and returns toward its initial baseline during deceleration. The signal amplitude increases with rotational velocity, reflecting the increase in centrifugal loading. Larger signal variations are also observed for the 50 mm flexure configuration than for the 30 mm configuration, consistent with the lower stiffness of the longer flexure.

Some discrepancies can nevertheless be observed between the ideal motor S-curve profile and the measured Hall-effect sensor outputs. In particular, the plateau region appears longer than expected. While the theoretical motor profile includes a plateau of approximately 1 s, the measured signals often exhibit longer plateau durations. Small oscillations also remain visible after Fourier filtering. These differences indicate that the measured response does not perfectly reproduce the theoretical motion profile.

Hall-effect sensor calibration

A calibration procedure was performed to relate the Hall-effect sensor output to the displacement of the moving carriage. The sensor output is proportional to the magnetic flux density generated by the permanent magnet, which decreases nonlinearly with distance according to

$$B(z) = \frac{B_r}{2} \left[\frac{z + L}{\sqrt{R^2 + (z + L)^2}} - \frac{z}{\sqrt{R^2 + z^2}} \right], \quad (4)$$

where B_r is the remanent flux density, R the magnet radius, L its length, and z the sensor-to-magnet distance. As a result, the relationship between measured voltage and mechanical displacement is inherently nonlinear, making the calibration a critical step for quantitative displacement reconstruction. The calibration curve is presented in Appendix.

Although the calibration provides a first estimate of the displacement sensitivity, it was established from a limited number of manually acquired points and remains specific to the sensor-magnet assembly used in this work. Consequently, the resulting displacement estimates should be interpreted as approximate values. The camera-based measurements are therefore used as the primary reference for quantitative displacement evaluation.

Camera-based displacement measurements

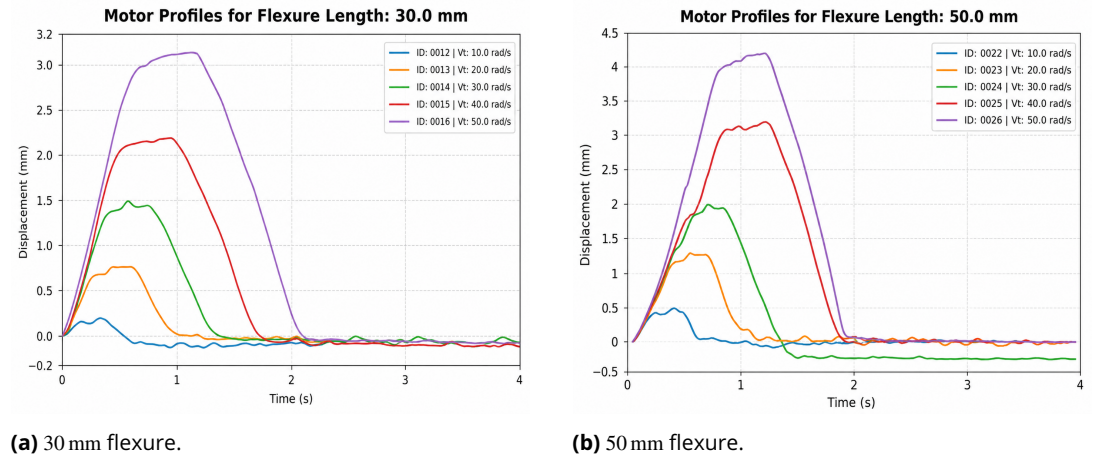


Figure 5. Camera-based displacement measurements obtained for the 30 mm and 50 mm flexure configurations under target angular velocities ranging from 10 rad s^{-1} , 20 rad s^{-1} , 30 rad s^{-1} , 40 rad s^{-1} and 50 rad s^{-1} .

The camera tracking method was used as a reference measurement technique to quantify the displacement of the moving carriage during the rotation experiments. Figure 5 presents the displacement responses obtained for the 30 mm and 50 mm flexure configurations over the investigated velocity range.

For both flexure lengths, the displacement amplitude increases with the imposed motor velocity. For the 30 mm flexure, the maximum displacement increases from approximately 0.6 mm at 10 rad s^{-1} to approximately 3.1 mm at 50 rad s^{-1} . For the 50 mm flexure, the

maximum displacement increases from approximately 0.5 mm to approximately 4.1 mm over the same velocity range. The larger displacements obtained with the longer flexures are consistent with their lower stiffness and therefore greater compliance under centrifugal loading.

The measured displacement profiles exhibit a progressive increase during the acceleration phase, followed by a plateau region corresponding to the constant-velocity stage and a smooth return towards the initial position during deceleration. This behaviour is consistent with the expected response of the flexure-guided carriage subjected to an increasing centrifugal force.

Since the centrifugal force is proportional to the square of the angular velocity according to $F^\Omega = mR\Omega^2$, the displacement evolution is not expected to follow the motor velocity profile linearly. Instead, the measured displacement increases more rapidly as the target angular velocity is approached and decreases progressively during deceleration. This nonlinear behaviour is clearly visible in both flexure configurations.

No significant overshoot or instability is observed during the acceleration or return phases, indicating that the flexure-guiding system remains stable throughout the investigated operating range.

In order to directly compare the Hall-effect sensor measurements with the camera-based displacement measurements and the imposed motor profile, the displacement curves were normalized with respect to their acquisition duration. The normalized time axis was then multiplied by the theoretical motor cycle duration associated with each target velocity. This reconstruction procedure allows all datasets to be represented on the same theoretical motor time scale while preserving the original displacement amplitudes.

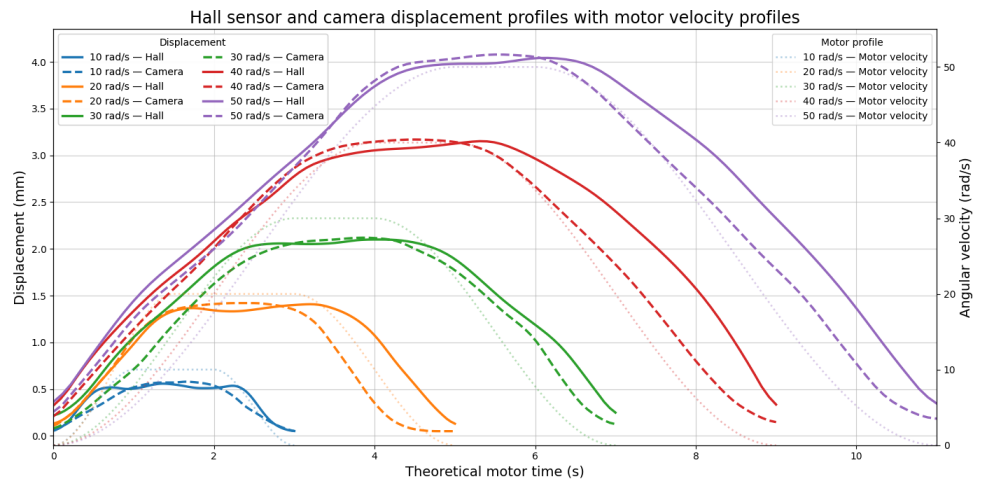


Figure 6. Comparison between the theoretical motor velocity profiles, the camera-based displacement measurements and the Hall-effect sensor measurements for the 50 mm flexure configuration. The displacement curves were first normalized with respect to their acquisition duration and subsequently reconstructed using the theoretical motor cycle duration corresponding to each target velocity.

A good agreement is observed between the Hall-effect sensor measurements and the camera-based displacement measurements over the entire velocity range. Both meth-

ods capture the same overall evolution of the displacement, including the progressive increase during acceleration, the plateau region near maximum displacement and the smooth decrease during deceleration.

The peak displacement values obtained from both measurement techniques are also in close agreement. At 50 rad s^{-1} , the Hall-effect sensor predicts a maximum displacement of approximately 4.03 mm, while the camera measurement yields approximately 4.08 mm, corresponding to a relative difference of approximately 1.2%. Similarly, at 40 rad s^{-1} , both methods predict peak displacements close to 3.1 mm. Comparable agreement is observed throughout the investigated velocity range.

The main differences between the two measurement techniques appear in the detailed temporal evolution of the displacement profiles. The Hall-effect sensor measurements generally exhibit smoother plateau regions and slight differences during the deceleration phase. Nevertheless, the overall shape, displacement amplitude and velocity-dependent trends remain highly consistent between both methods.

These results demonstrate that the Hall-effect sensor is capable of reproducing the displacement evolution measured by the optical tracking system while providing a compact measurement solution suitable for integration directly on the rotating frame. However, the calibration procedure remains dependent on the specific sensor-magnet assembly and would benefit from a more systematic characterization using a larger number of calibration points and multiple sensor configurations.

Dynamic response and natural frequency of the flexures

The dynamic behaviour of the flexure system was first investigated through free oscillation experiments. After applying an initial displacement to the moving carriage, the displacement evolution was recorded using the imaging method described previously.

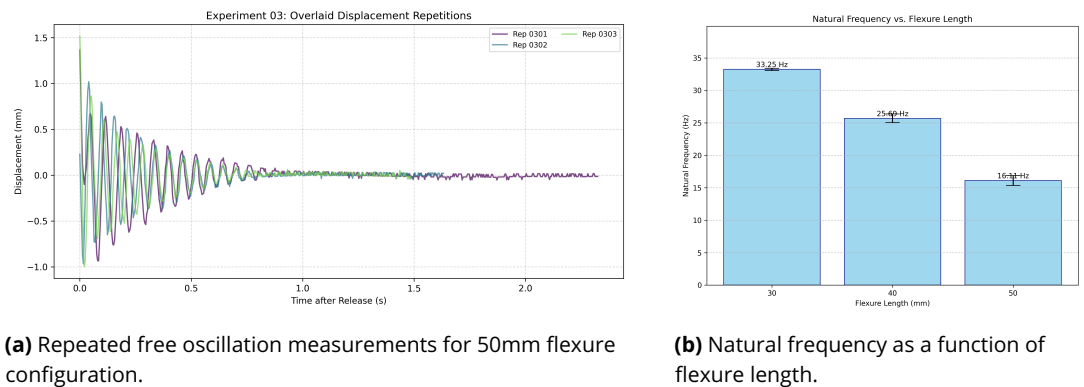


Figure 7. Dynamic response of the flexure system. (a) Overlaid displacement repetitions showing the damped oscillatory behaviour of the moving carriage. (b) Measured natural frequencies for different flexure lengths.

Figure 7(a) shows three repeated displacement measurements obtained for the same 50 mm flexure configuration. The oscillatory response exhibits a progressively decreasing amplitude characteristic of a damped mechanical system. The oscillation frequency was

then extracted for flexures of different lengths in order to characterise the influence of stiffness on the dynamic response of the system. The resulting natural frequencies are presented in Figure 7(b) for three flexure lengths.

The results show a clear decrease in natural frequency as the flexure length increases. The 30 mm flexure exhibits a natural frequency of approximately 33.25 Hz, while the 50 mm flexure decreases to approximately 16.11 Hz. This behaviour is consistent with Euler–Bernoulli beam theory, since increasing the flexure length reduces the structural stiffness according to $k \propto 1/\ell^3$.

These results confirm that the flexure geometry strongly influences the dynamic response of the system and therefore the displacement amplitudes reached during rotational loading. The natural frequency characterisation also provides important information for selecting operating conditions that avoid unwanted resonance effects during testing experiments.

Conclusion

This work presented the development and experimental validation of a centrifugal testing apparatus combining flexure-guided test stations with optical and Hall-effect sensor displacement measurements.

The camera-based measurements successfully captured the displacement response of the system and showed good agreement with the imposed motor driving profiles. The results confirmed the expected influence of motor velocity and flexure length on the displacement amplitude, validating both the measurement method and the overall behaviour of the experimental setup.

The Hall-effect sensor measurements captured the same general trends but did not fully reproduce the displacement profiles observed with the camera. In particular, differences in the plateau regions and the need for temporal corrections highlighted limitations in the current calibration and acquisition procedure. Consequently, the Hall-effect sensor approach cannot yet be considered fully validated and requires further investigation.

Future work should focus on improving the Hall-effect sensor calibration using an instrumented testing machine capable of continuously synchronizing displacement and voltage measurements. The optical tracking system could also be improved by integrating the lighting directly into the rotating frame. Finally, the methodology should be extended to tests involving actual specimens in order to characterize their mechanical behaviour and failure under centrifugal loading.

Annexes

Hall-effect sensor calibration

A preliminary calibration of the Hall-effect sensor was performed prior to the rotation experiments by positioning the permanent magnet at known distances from the sensor and recording the corresponding raw Arduino output.

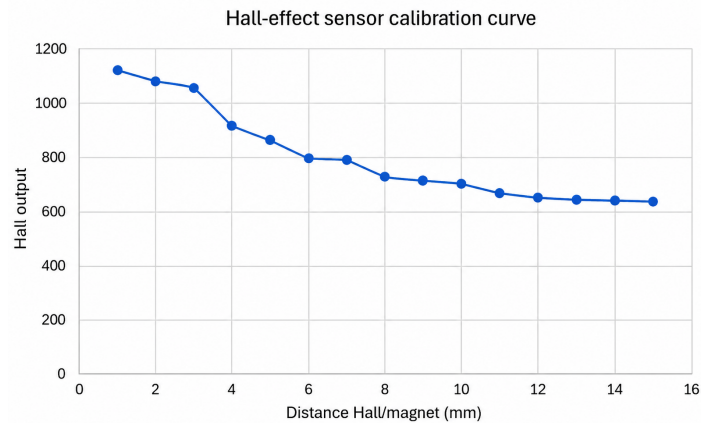


Figure 8. Preliminary Hall-effect sensor calibration curve. The raw Arduino output signal is plotted as a function of the sensor-to-magnet distance.

Figure 8 shows the resulting calibration curve. As expected, the sensor response is non-linear, with a higher sensitivity at short distances and a saturation at larger distances.

The calibration was used to obtain approximate displacement estimates from the Hall-effect sensor measurements. However, the limited number of manually acquired calibration points introduces uncertainty. A more rigorous calibration could be achieved in future work using an instrumented testing machine capable of continuously synchronizing displacement and sensor output measurements.

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